Discussion week 6

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Nov 4, 2021

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I.

1. During the spin cycle of a washing machine, the clothes stick to the outer wall of the barrel as it spins at a rate as high as 1800 revolutions per minute. The radius of the barrel is 26 cm.
   a. Determine the speed of the clothes (in m/s) which are located on the wall of the spin barrel.
   b. Determine the acceleration of the clothes.
   c. If the washing machine barrel takes five second to reach its maximum spin rate, what is the angular acceleration of the clothes during this time?

I.

a. Determine the speed of the clothes (in m/s) which are located on the wall of the spin barrel.

$$v = \omega r = 1800 rev/60s \times 2\pi \times 0.26m = 49.0m/s$$

b. Determine the acceleration of the clothes.

$$a = \frac{v^2}{r} = 49^2/0.26\ m/s^2 \approx 9238m/s^2$$

d. If the washing machine barrel takes five second to reach its maximum spin rate, what is the angular acceleration of the clothes during this time?

$$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{60\pi}{5\ s} = 12\pi\ rad/s$$

2.

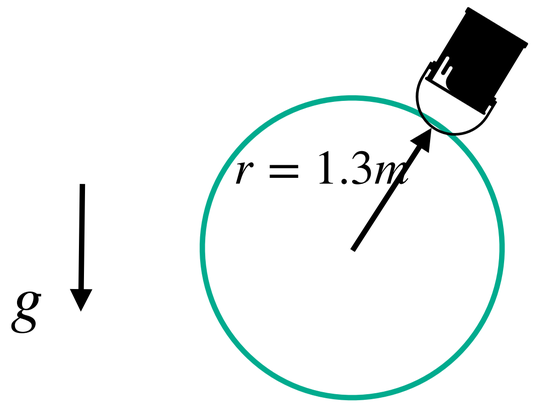
2. Alexis is in her Toyota Camry and trying to make a turn off an expressway at 19.0 m/s. The turning radius of the level curve is 35.0 m. Her car has a mass of 1240 kg. Determine the acceleration, net force and minimum value of the coefficient of friction which is required to keep the car on the road.

$$a = \frac{v^2}{r} = \frac{19^2}{35} m/s^2 = 10.3 m/s^2$$

$$F_{net} = ma = 1240 \times 10.3 N = 12789.7 N$$

$$f_s^{max} = \mu_s mg \geq F_{net} \rightarrow \mu_s \geq a/g \approx 1.05$$

3.



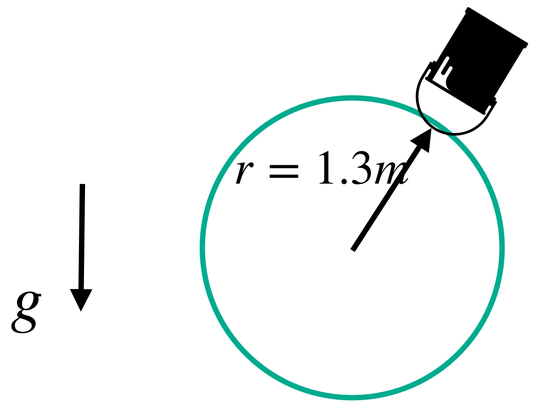
3. In an effort to rev up his class, Mr. H does a demonstration with a bucket of water tied to a 1.3-meter long string. The bucket and water have a mass of 1.8 kg. Mr. H whirls the bucket in a vertical circle such that it has a speed of 3.9 m/s at the top of the loop and 6.4 m/s at the bottom of the loop.

a. Determine the acceleration of the bucket at each location.

b. Determine the net force experienced by the bucket at each location.

c. Draw a free body diagram for the bucket for each location and determine the tension force in the string for the two locations.

3.



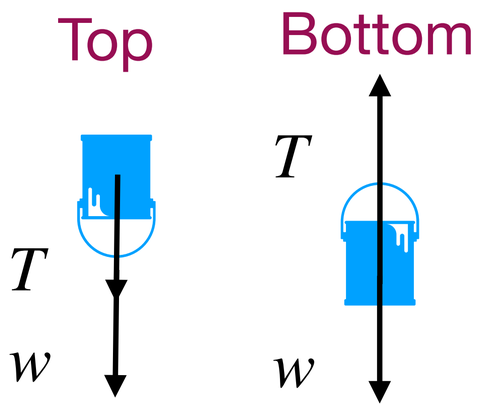
a. Determine the acceleration of the bucket at each location.

$$a_{top} = \frac{3.9^2}{1.3} m/s^2 = 11.7 m/s^2 \qquad a_{top} = \frac{6.4^2}{1.3} m/s^2 = 31.5 m/s^2$$

b. Determine the net force experienced by the bucket at each location.

$$F_{net} = ma$$

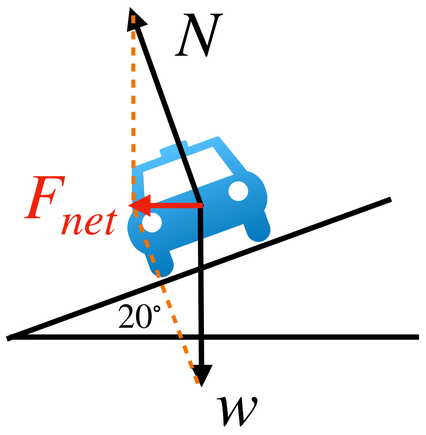
c. Draw a free body diagram for the bucket for each location and determine the tension force in the string for the two locations.



Top: $T_{top} + w = m\frac{v_{top}^2}{r}$ Bottom: $T_{bot} - w = m\frac{v_{bot}^2}{r}$

4.

4. Curvy roads in places that get ice (ie, not Riverside) are sometimes "banked," meaning they are sloped toward the center of curvature. If banked at the correct angle, cars don't need *any* friction to make it around the curve (at the correct speed). How fast would a car have to go around a curve with a radius of 100m and a band of 20 degrees?



$$F_{net} = mg\tan 20^\circ = m\frac{v^2}{r}$$

$$v = \sqrt{gr\tan 20^\circ} = \sqrt{9.8 \times 100 \times \tan 20^\circ} \approx 18.9 m/s^2$$